

Fig. 6: Alter Nrg plasma gasification system using WPC PGVR

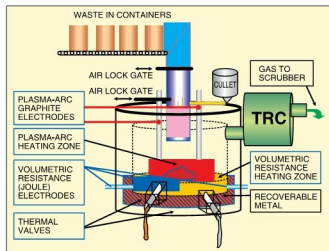


Fig. 7: Schematic of PEM process vessel with side-entry AC electrodes

using multiple graphite electrodes, joule (resistance) heating using glass melter technology and superheated steam. This combination results in a stable and highly controllable treatment system.

The plasma operates at temperatures from 3000°C to 10,000°C in an oxygen-free environment with the presence of superheated steam ensuring that pyrolysis and steam reforming reactions dominate. A high-efficiency scrubber is used to remove volatile metals and other pollutants from the hydrogen-rich product gas (100 BTU/scf or more), a portion of which may be recycled to provide power

and the composition of the waste.

IET is more interested in treating radioactive, hazardous, industrial, municipal, tire, incinerator ash and medical waste streams. It has sold several commercial PEM units throughout the world to process a wide range of waste materials. For details, visit 'www.inentec.com/commercial.html'.

The Solena Group. The Solena Group, a strategic partner of Westinghouse, uses the integrated plasma gasification combined cycle (IPGCC). It is a unique process that uses a patented plasma gasification vitrification (PGV) technology for conversion of low-value fuels/feed stocks

to the PEM and the other portion used to generate electricity.

Metals in the waste are recovered. The glassy aggregate is also recovered and may be recycled as road building, blasting grit or construction material. Volume reductions are up to 98 per cent depending on how the process is run

(such as coal waste, petroleum coke, bio-mass and different waste forms including municipal solid waste) into a low-heating-value synthesis or syngas containing carbon monoxide and hydrogen gases. This gas is then employed as a primary fuel for a gas turbine.

The IPGCC can also be viewed as a two-stage processor of an opportunity feedstock. The feedstock is first gasified (partially oxidised) in a gasifier using high-temperature plasma heating systems at atmospheric pressure. The gas is then cleaned and combusted (completely oxidised) in the gas turbine to produce electricity.

The PGV reactor, in essence, cleans the solid/liquid and/or dirty fuels/feed stocks by converting them into a valuable byproduct fuel gas or syngas to power a gas turbine.

The IPGCC system generally consists of four separate processes: feedstock handling, plasma gasification and vitrification, gas cooling and clean-up, and gas turbine combined cycle.

The IPGCC system offers significant advantages over standard IGCC. By applying the PGV technology for gasifying the feedstock, the plasma heating system provides an independent and extremely powerful heating